Thermal expansion coefficient

We are interested in

$$\alpha \equiv \frac{1}{V}\frac{\partial V}{\partial T}\bigg|_p \,, \tag{1}$$

where V denotes the volume and p the pressure. The relevant deformation is defined by the affine transformations of coordinates $\boldsymbol{x} \to \mathcal{H}\cdot\boldsymbol{x}$ as

$$\mathcal{H} = \begin{pmatrix} e^\eta & 0 & 0 \\ 0 & e^\eta & 0 \\ 0 & 0 & e^\eta \end{pmatrix} , \tag{2}$$

and the pressure is

$$p = -\frac{1}{V\text{đ}}\frac{\partial U}{\partial \eta} \,, \tag{3}$$

where đ denotes the dimension of space. Using the relation

$$\left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_V \left(\frac{\partial p}{\partial V}\right)_T = -1 \,, \tag{4}$$

one obtains

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p = -\frac{1}{V}\frac{(\partial p/\partial T)_V}{(\partial p/\partial V)_T} \,. \tag{5}$$

We want to calculate α in inherent states of various glasses. Notice that the athermal ($T \to 0$) bulk modulus is defined as

$$K \equiv -V\frac{\mathcal{D}p}{\mathcal{D}V} = \frac{1}{V\text{đ}^2}\frac{\mathcal{D}^2 U}{\mathcal{D}\eta^2} + p \,, \tag{6}$$

where $\mathcal{D}/\mathcal{D}\eta$ denotes the zero-temperature limit of derivatives at constant temperature, which are derivatives at constant (zero) net-forces $\partial U/\partial \boldsymbol{x} = \boldsymbol{0}$, in discussion of this arXiv. We can therefore conclude that

$$\alpha = \frac{1}{K}\left(\frac{\partial p}{\partial T}\right)_V \,. \tag{7}$$

From PNAS 2017 we know that

$$2\frac{\partial p}{\partial T}\bigg|_V = \frac{\partial^2 p}{\partial \boldsymbol{x}\partial \boldsymbol{x}} : \boldsymbol{\mathcal{H}}^{-1} - \frac{\partial p}{\partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \boldsymbol{\mathcal{U}}''' : \boldsymbol{\mathcal{H}}^{-1} \,. \tag{8}$$

$$\boldsymbol{\mathcal{H}} = \sum_\omega \omega^2 \Psi_\omega \otimes \Psi_\omega \quad \text{and so} \quad \boldsymbol{\mathcal{H}}^{-1} = \sum_\omega \frac{1}{\omega^2}\,\Psi_\omega \otimes \Psi_\omega .$$

To derive microscopic expressions, we first work out how pairwise distances $r_{ij} \equiv |\boldsymbol{x}_j - \boldsymbol{x}_i| = |\boldsymbol{x}_{ij}|$ vary with respect to coordinates and deformations. Here are the relevant derivatives

$$\begin{aligned} \frac{\partial \boldsymbol{x}_{ij}}{\partial \eta} &= \boldsymbol{x}_{ij} \,, \\ \frac{\partial r_{ij}}{\partial \eta} &= r_{ij} \,, \\ \frac{\partial r_{ij}}{\partial \boldsymbol{x}_k} &= (\delta_{jk} - \delta_{ik})\frac{\boldsymbol{x}_{ij}}{r_{ij}} \,, \\ \frac{\partial \boldsymbol{x}_{ij}}{\partial \boldsymbol{x}_k} &= (\delta_{jk} - \delta_{ik})\boldsymbol{\mathcal{I}} \,, \end{aligned}$$

The energy is assumed to have the form

$$U = \sum_{i<j} \varphi_{ij}(r_{ij}) \,. \tag{9}$$

In what follows we use the following shorthand notations

$$\begin{aligned} \frac{\varphi'}{r} &\equiv [1\text{st}]\,, \\ \frac{\varphi''}{r^2} - \frac{\varphi'}{r^3} &\equiv [2\text{nd}]\,, \\ \frac{\varphi'''}{r^3} - 3\frac{\varphi''}{r^4} + 3\frac{\varphi'}{r^5} &\equiv [3\text{rd}]\,, \end{aligned}$$

and these satisfy the relations

$$\begin{aligned} \frac{\partial [1\text{st}]}{\partial r} &= [2\text{nd}] \times r\,, \\ \frac{\partial\,[2\text{nd}]}{\partial r} &= [3\text{nd}] \times r\,, \end{aligned}$$

We now start with derivatives with respect to coordinates only, namely

$$\frac{\partial U}{\partial \boldsymbol{x}_k} = \sum_{i<j} (\delta_{jk} - \delta_{ik}) \frac{\varphi'_{ij}}{r_{ij}} \boldsymbol{x}_{ij} = \sum_{i<j} (\delta_{jk} - \delta_{ik}) [1\text{st}] \boldsymbol{x}_{ij}\,. \tag{10}$$

$$\frac{\partial^2 U}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_\ell} = \sum_{i<j} (\delta_{jk} - \delta_{ik})(\delta_{j\ell} - \delta_{i\ell}) \Big[[2\text{nd}] \boldsymbol{x}_{ij} \boldsymbol{x}_{ij} + [1\text{st}] \boldsymbol{\mathcal{I}} \Big]\,. \tag{11}$$

$$\frac{\partial^3 U}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_\ell \partial \boldsymbol{x}_m} = \sum_{i<j} (\delta_{jk} - \delta_{ik})(\delta_{j\ell} - \delta_{i\ell})(\delta_{jm} - \delta_{im}) \Big[[3\text{rd}] \boldsymbol{x}_{ij} \boldsymbol{x}_{ij} \boldsymbol{x}_{ij} + \text{`3'} [2\text{nd}] \boldsymbol{\mathcal{I}} \boldsymbol{x}_{ij} \Big]\,. \tag{12}$$

Derivatives with respect to deformations.

$$\frac{\partial U}{\partial \eta} = \sum_{i<j} \varphi'_{ij} r_{ij} = \sum_{i<j} [1\text{st}] r_{ij}^2\,, \tag{13}$$

Mixed derivatives:

$$\frac{\partial^2 U}{\partial \eta \partial \boldsymbol{x}_k} = \sum_{i<j} (\delta_{jk} - \delta_{ik}) \left(\varphi''_{ij} + \frac{\varphi'_{ij}}{r_{ij}} \right) \boldsymbol{x}_{ij} = \sum_{i<j} (\delta_{jk} - \delta_{ik}) \Big[[2\text{nd}] r_{ij}^2 \boldsymbol{x}_{ij} + 2[1\text{st}] \boldsymbol{x}_{ij} \Big]\,. \tag{14}$$

$$\frac{\partial^3 U}{\partial \eta \partial \boldsymbol{x} \partial \boldsymbol{x}} = \sum_{i<j} (\delta_{jk} - \delta_{ik})(\delta_{j\ell} - \delta_{i\ell}) \Big[[3\text{rd}] r_{ij}^2 \boldsymbol{x}_{ij} \boldsymbol{x}_{ij} + [2\text{nd}] (4 \boldsymbol{x}_{ij} \boldsymbol{x}_{ij} + r_{ij}^2 \boldsymbol{\mathcal{I}}) + 2[1\text{st}] \boldsymbol{\mathcal{I}} \Big] \tag{15}$$

Evaluation of $(\partial p / \partial T)_V$

Recall that

$$p = -\frac{1}{V đ} \frac{\partial U}{\partial \eta}\,. \tag{16}$$

Therefore

$$\frac{\partial p}{\partial \boldsymbol{x}_k} = -\frac{1}{V đ} \frac{\partial^2 U}{\partial \eta \partial \boldsymbol{x}_k}\,, \tag{17}$$

and

$$\frac{\partial^2 p}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_\ell} = -\frac{1}{V đ} \frac{\partial^3 U}{\partial \eta \partial \boldsymbol{x}_k \partial \boldsymbol{x}_\ell}\,. \tag{18}$$

Using the expressions derived above we obtain

$$\frac{\partial p}{\partial \boldsymbol{x}_k} = -\frac{1}{V đ} \sum_{i<j} (\delta_{jk} - \delta_{ik}) \Big[[2\text{nd}] r_{ij}^2 \boldsymbol{x}_{ij} + 2[1\text{st}] \boldsymbol{x}_{ij} \Big]\,. \tag{19}$$

and

$$\frac{\partial^2 p}{\partial \boldsymbol{x}_k \partial \boldsymbol{x}_\ell} = -\frac{1}{V đ} \sum_{i<j} (\delta_{jk} - \delta_{ik})(\delta_{j\ell} - \delta_{i\ell}) \Big[[3\text{rd}] r_{ij}^2 \boldsymbol{x}_{ij} \boldsymbol{x}_{ij} + [2\text{nd}] (4 \boldsymbol{x}_{ij} \boldsymbol{x}_{ij} + r_{ij}^2 \boldsymbol{\mathcal{I}}) + 2[1\text{st}] \boldsymbol{\mathcal{I}} \Big] \tag{20}$$

Low-Temperature Expansion

From PNAS 2017 we use

$$2\frac{\partial p}{\partial T}\bigg|_V = \frac{\partial^2 p}{\partial \boldsymbol{x}\partial \boldsymbol{x}} : \boldsymbol{\mathcal{H}}^{-1} - \frac{\partial p}{\partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \boldsymbol{\mathcal{U}}''' : \boldsymbol{\mathcal{H}}^{-1}\,. \tag{21}$$

Substituting the results above yields

$$2\frac{\partial p}{\partial T}\bigg|_V = -\frac{1}{V\đ}\frac{\partial^3 U}{\partial \eta \partial \boldsymbol{x}\partial \boldsymbol{x}} : \boldsymbol{\mathcal{H}}^{-1} + \frac{1}{V\đ}\frac{\partial^2 U}{\partial \eta \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \boldsymbol{\mathcal{U}}''' : \boldsymbol{\mathcal{H}}^{-1}\,. \tag{22}$$

Final Microscopic Expression for α

Using

$$\alpha = \frac{1}{K}\left(\frac{\partial p}{\partial T}\right)_V \tag{23}$$

we obtain

$$\boxed{\alpha = \frac{1}{2KV\đ}\left[-\frac{\partial^3 U}{\partial \eta \partial \boldsymbol{x}\partial \boldsymbol{x}} : \boldsymbol{\mathcal{H}}^{-1} + \frac{\partial^2 U}{\partial \eta \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \boldsymbol{\mathcal{U}}''' : \boldsymbol{\mathcal{H}}^{-1}\right]} \tag{24}$$

This expression contains two distinct contributions:

- A **harmonic contribution**
$$-\frac{1}{2KV\đ}\frac{\partial^3 U}{\partial \eta \partial \boldsymbol{x}\partial \boldsymbol{x}} : \boldsymbol{\mathcal{H}}^{-1}$$
arising from curvature of the pressure field.
- An **anharmonic contribution**
$$\frac{1}{2KV\đ}\frac{\partial^2 U}{\partial \eta \partial \boldsymbol{x}} \cdot \boldsymbol{\mathcal{H}}^{-1} \cdot \boldsymbol{\mathcal{U}}''' : \boldsymbol{\mathcal{H}}^{-1}$$
which couples mismatch forces to the third-order elastic tensor.

Importantly, mechanical equilibrium

$$\frac{\partial U}{\partial \boldsymbol{x}} = \boldsymbol{0}$$

does *not* eliminate the harmonic term. Therefore both terms contribute at leading order.

Conclusion

The isobaric thermal expansion coefficient in inherent states is determined by the temperature derivative of the pressure field at fixed volume. Its microscopic expression consists of two distinct contributions: a harmonic curvature term and an anharmonic coupling term. Both scale as ω_ℓ^{-4} in the normal mode representation, implying dominance by soft modes in amorphous solids.